

Climate change and renewable energy

Glossary

anthropogenic greenhouse effect

> This can also be called the enhanced greenhouse effect. That part of the greenhouse effect that can be attributed to human action (for example, burning fossil fuels, deforestation). See greenhouse effect.

battery

> A device for storing electrical energy in a chemical form and supplying electric current. It usually consists of one or more electrical cells connected together in series.

battery electric vehicles

> (BEVs) A vehicle that runs on electricity drawn from a battery (as opposed to a fuel cell).

biofuels

> These are fuels produced directly or indirectly from living organic material – biomass, including plant materials and animal waste.

biomass

> Materials such as wood, plant, animal waste, etc. that (unlike fossil fuels) were living matter relatively recently. *Traditional biomass* refers to biomass resources that are not formally traded, in contrast to new biomass.

black hydrogen

> Hydrogen produced from coal. That from other sources is assigned different ‘colours’.

blue hydrogen

> Hydrogen produced from natural gas but using carbon capture and storage to reduce the associated CO_2 emissions.

carbohydrates

> (CHOs) Collective name for many organic compounds made from carbon, hydrogen and oxygen, e.g. starch, sugars, cellulose; one of the major food groups.

carbon capture, utilisation and storage (CCUS)

> Techniques to separate carbon from fossil fuels when they are used to provide energy for any purpose, but particularly for power stations and energy-intensive industries. Carbon can be removed from the fuel before combustion or from the waste gases after combustion. In both cases it is extracted in the form of carbon dioxide, which is then transported to a storage site, such as a depleted oil or gas reservoir, for long-term burial.

carbon cycle

> The movement of carbon between reservoirs such as the atmosphere, organisms, soil, and some rocks.

carbon dioxide

A naturally occurring chemical compound consisting of carbon and oxygen. A molecule of carbon dioxide consists of one atom of carbon and two atoms of oxygen with a chemical formula of CO_2. Carbon dioxide is produced in respiration and from the burning of carbon-intensive fossil fuels and consumed in photosynthesis. Carbon dioxide is the most important greenhouse gas released as a result of human activity.

concentrated solar power (CSP)

The use of sunlight to generate high-temperature steam by means of an array of mirrors. The steam then drives a turbine in the usual way.

COP

Conference of Parties – the yearly conference where the signatories of the United Nations Framework Convention on Climate Change (UNFCCC) meet.

electrolysis

The chemical decomposition of a liquid or solution that occurs during the conduction of electricity.

energy services

The ultimate aims for which energy systems are built: warm homes, cooked food, illumination, mobility and manufactured articles.

environment

The surroundings and influences on living things including humans.

excreta

The compounds disposed of by organisms in liquid, solid or gaseous form that are no longer needed or are harmful.

fertiliser

Any material used by humans as a source of plant nutrients.

fossil fuel

Buried fuels derived from past living plant and animal materials that have been modified and buried by geological processes, for example coal, oil or gas. The burning of fossil fuels causes carbon dioxide emissions and is the major cause of climate change.

fuel cell

A device for producing an electric current by means of what is essentially the reverse process to electrolysis – combining two gases (typically hydrogen and oxygen) to produce electricity.

gigatonne

Giga is the prefix for 10^9 so one gigatonne is the same as 1000 megatonnes or a billion tonnes.

global warming

The rising average world temperature, often measured by the global mean surface temperature. Within this average it will still be possible for countries and regions to have colder weather.

global mean surface temperature (GMST)

This is a measure of the global average temperature of the Earth, measured near the surface. It is an average of (a) the sea surface temperature and (b) the air temperature measured 1.5 metres above ground, weighted for area.

Global Warming Potential

Global Warming Potential (GWP) is a relative measure of how much heat a greenhouse gas traps in the atmosphere. It compares the amount of heat trapped by a certain mass of the gas in question to the amount of heat trapped by a similar mass of carbon dioxide.

greenhouse effect

Natural mechanism, now modified by human activity, whereby solar radiation is trapped by carbon dioxide and other greenhouse gases in the Earth's atmosphere, similar to the way in which heat is trapped by glass in a greenhouse.

greenhouse gases

(GHG) Gases, including carbon dioxide, water vapour, methane and nitrous oxide, that interact with infra-red radiation and when present in the atmosphere have the effect of warming the global climate. Without naturally occurring greenhouse gases the Earth's temperature would be several tens of degrees Celsius colder than it is now (and life would not have evolved in its current form).

green hydrogen

Hydrogen that has been produced from renewable energy sources with minimal associated carbon dioxide emissions.

grey hydrogen

Hydrogen produced from natural gas, but without any use of carbon capture and storage to reduce the consequent CO_2 emissions.

heat pump

A device that 'pumps' heat from a cooler region into a warmer one, thus providing either warming or cooling to a living space. In an air-source heat pump (ASHP) the heat is taken directly from the air, while in a ground-source heat pump (GSHP) it comes from the soil but indirectly from the air.

hydroelectric power

Power harnessed from converting the energy coming of running water into electrical energy.

IEA

International Energy Agency

infra-red radiation

Electromagnetic radiation with a wavelength longer than visible light.

IPCC

The Intergovernmental Panel on Climate Change was established in 1988 by the World Meteorological Organisation (WMO) and the United Nations Environment Programme (UNEP). Its purpose is to provide scientific and technical advice to the United Nations Framework Convention on Climate Change (UNFCC) and assess current information on climate change, its impacts and how to mitigate (reduce) the effects and/or to adapt to it. It produces authoritative reports on these matters approximately every five years.

Keeling Curve

The graphical record of the rise of carbon dioxide concentration in the atmosphere from 1958 to the present from the Mauna Loa Observatory in Hawaii. It is named after Charles Keeling, the scientist who established the observations.

Kyoto Protocol

An international agreement stemming from the 1997 climate change conference held in the Japanese city of Kyoto. Under this protocol, developed countries (except the USA and Australia) in 2005 agreed a legally-binding obligation to reduce their overall emissions of greenhouse gases, contributing to climate change by 5.2% between 2008 and 2012, based on 1990 emissions. The agreement refers to a basket of the six main human-induced greenhouse gases, particularly carbon dioxide. Australia finally signed the treaty in 2007. In 2012, the Protocol was extended until 2020 to provide more time for international agreement on a successor commitment.

leaf litter

A habitat formed of dead and decaying leaves and other organic material.

LED

Light emitting diode (LED) lamps are based on the properties of materials called semiconductors. They have many advantages over incandescent light sources, including lower energy consumption, longer lifetime, improved physical robustness, and smaller size.

mass extinction

During the course of the history of life on Earth there have been several occasions when a large proportion of all species and many higher groupings of organisms have become extinct in a comparatively short geological time. The specific causes are not clear and may be different in each case, but the five or more occasions of mass extinction all took place when there were major changes to the Earth's systems: the oceans, ice, atmosphere, land and ecosystems. There is concern that human activity is now in danger of leading to another mass extinction.

methane

A hydrocarbon that is a constituent of natural gas. Also produced by decomposing organic waste such as domestic refuse and by digestion processes in animals. Methane is a potent greenhouse gas and has a chemical formula of CH_4.

micro-organisms

Very small (usually single-celled) organisms only visible with a microscope, e.g. bacteria.

nationally determined contributions (NDCs)

The climate pledges (in terms of reductions in greenhouse gases) that countries made at the United Nations Climate Change Conference held in Paris, France in December 2015. Previously known as intended nationally determined contributions (INDCs).

net zero

Net zero emissions (also referred to as zero emissions or carbon neutral) are achieved when greenhouse gas emissions are balanced by technologies or practices that remove emissions from the atmosphere, resulting in a sum total of zero.

nitrous oxide

A colourless gas used as an anaesthetic. It occurs naturally in the atmosphere and is mainly formed by soil bacteria decomposing nitrogen-containing material. Agricultural practices affect its release into the atmosphere where it is a greenhouse gas (N_2O)

photosynthesis

The process in which green plants (and a few other organisms including plankton) in the presence of sunlight use carbon dioxide and water to produce carbohydrates and oxygen.

phytoplankton

Very small plants (many are microscopic) that drift passively in the sea or in lakes.

pink hydrogen

Hydrogen produced by the electrolysis of water using nuclear electricity.

primary energy

The total energy content of an energy resource before that energy is extracted/transformed/processed. Primary energy is the starting point for an energy transformation system: the incoming 'amount we have to work with' before it is acted upon by power stations.

renewable energy

Energy based on a naturally occurring and replenished source, such as sunlight, wind, biomass, waves or tides.

respiration

The process whereby living organisms convert stored organic carbon compounds into carbon dioxide and other compounds, in the process releasing energy and taking up oxygen.

ruminants

Even-toed hoofed mammals with a complex 3- or 4-chamber stomach. Ruminants are herbivores (only eat plants), and their digestion creates methane which is exhaled or belched. Ruminants include cattle, sheep, antelopes, deer and giraffes.

sequestration

The very long-term storage of CO_2 or solid carbon in isolation from the atmosphere.

spectrum

> The electromagnetic spectrum is the range of wavelengths or frequencies over which electromagnetic radiation extends.

ultraviolet

> (UV) This is an electromagnetic radiation with a wavelength shorter than that of infra-red radiation, visible light but longer than than that of X-rays.

water vapour

> Water in the state of a vapour (similar to a gas). It is present in the atmosphere in variable amounts, typically about 3% of the dry atmosphere and plays an important role in the water cycle, in cloud formation and controlling the distribution of heat through the atmosphere.

zooplankton

> Very small animals (many are microscopic) that drift passively in the sea or in lakes.

Previous

Acknowledgements

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